This document is intended as a guide for the senior project team to assess its performance in a number of dimensions. Add additional items that you feel are appropriate.

This self-assessment will be one of multiple elements that your faculty coach uses to arrive at an assessment of the team's performance for this second term. The other elements that the faculty coach will use include: direct observation of the team, team peer evaluations, reviews by other faculty during the project presentation, sponsor evaluation, and project deliverables. These self-assessments will also be used as part of the SE program's accreditation and curriculum improvement efforts.

To complete this self-assessment the team should carefully consider each of the questions and provide an honest evaluation of the team's performance.

Product

1. Did the team prepare all the documentation artifacts requested by your faculty coach and sponsor? Were these documents carefully inspected prior to delivery? How would you assess the quality of the document artifacts?

Yes, the team prepared all documentation artifacts requested by the coach and sponsor, they were carefully inspected by the team and coach prior to delivery to the sponsor. The quality of the artifacts is up to professional standards.

2. How well did the team elicit the requirements? What approaches were used to elicit the requirements? Were key requirements missed? What methodology was used to document and validate the project requirements?

The team elicited requirements well, by meeting with the sponsor many times and clarifying their expectations. Additionally, wireframes, mockups, and documents listing the proposed flow and use cases were used to ensure as few defects as possible made it out of the elicitation phase. The key requirement of a way to manage devices was missed from initial elicitation, but corrected later on.

3. Did the team explore the entire design space before arriving at a final design? Have there been many errors found in the design? Was it necessary to make major changes to any part of the design? What were the reasons for the change?

The team explored the vast majority of the design space before arriving at a final design, which took a long time. Exploring everything would've taken too long especially because certain decisions had a deadline. No errors have been found in the core design, only enhancements. Major changes had to be made to the inherited architecture, as it severely hampered performance and security.

4. How has the development and implementation progressed? What percentage of the product do you estimate was completed? Is the team providing the documentation within the implementation artifacts?

Development and implementation progressed smoothly, though longer than expected. We expected ~120% of the original project requirements were completed, as all of MVP was established as well as a few enhancements. Documentation is provided separately from implementation artifacts.

5. What was the team's testing strategy? Did the team develop a test plan? If so, was it followed? Did the team perform unit testing? Did the team use any test frameworks, such as JUnit? What are the testing results? Were any major defects found during system testing? If so, were they fixed? Did the team do regression testing?

The team tests at every stage of the SDLC, from elicitation all the way to acceptance testing. A feature starts as an idea from the team or sponsor, gets clarified and added to the requirements, flow, and use case document, and is added to the backlog. Once pulled from the backlog, the feature is designed, implemented, and tested via unit/component tests, integration tests, and finally UI tests. Once all tests pass and there is good enough coverage, it is sent for user acceptance testing and from there production and demo. Few defects were found, no major defects. All were fixed. The team does regular regression testing.

6. Products need to be designed within guidelines and constraints appropriate for each project. It is also important to consider the impacts of the products that are designed. In the following categories discuss the constraints and impacts that have a bearing on your project. Note that all of these categories may not have bearing on your project but your project is probably affected by many of them in ways that you may not think of regularly.
    a. Economic issues

        The team is constrained by the resources of the sponsor organization, it cannot use all of the features available to them as they cost money. This impacts how fast certain milestones can be met, and often incurs a cost in time.

    b. Environmental issues

        The team must take outages into account caused by unstable environmental conditions. This is a constraint of connectivity that is felt by both hardware and software. The hardware must be easy to provision, block, and test as a result.

    c. Social issues

        As this software is going to be going into public areas, it's important the team take its target audience into account. Hikers and trail managers will be using this platform, and that audience has specific social wants and needs the team must take into account when designing the look and feel of the platform.

- Sponsor: #1 audience is Stewardships Managers, **then** hikers (not 50/50, 90/10).

d. Political issues

It is hoped that the software will be used and managed by the NYS DEC to make political decisions that impact the park, partly on data that they take from this platform. This impacts how the platform must be designed, maintained, and implemented. Additionally, it also imposes the constraint of maintaining correctness, even under adverse conditions.

e. Ethical issues

This platform may be used to drive political decision making. It is the platform's ethical responsibility to provide accurate data without bias.

f. Health and safety

N/A

g. Manufacturability

The manufacturability of devices is constrained by how easy it is to provision them through the website. This process must be quick and easy enough to provision many devices at cost, but not so easy as to be insecure.

h. Sustainability

The platform needs to be operational for many years into the future to be accurate and useful. Thus, the groundwork of the platform must be solid and well engineered to work for many years into the future with minimal refactoring and overhauls.

7. What industry or engineering standards was your project required to adhere to? Were these new standards that the team had to learn? Did your sponsor provide you support for understanding these standards? Did you have to educate your sponsor about these standards?

Industry and engineering standards for government run software are stringent as it pertains to user data and data sent over the network. Some of this was required to be learned by the team and sponsor, much through the team's independent research.

8. What standard software engineering practices did you follow? Did your sponsor specify any of these practices for you?

The team followed all standard software engineering practices: AGILE, regular standups, structured meetings, designs reviews, code reviews, change management, automated and manual testing, and the list goes on. Some standard practices were bypassed in the name of speed, but that can be chalked up to attempting to follow AGILE methodology. The sponsor specified the usage of AGILE, and the team agreed.

Process and Project

1. What was your process methodology? Was the process appropriate for the project? Did you follow the process or modify it as the project progressed? If you could repeat the project, what would you do differently?

The team follows an agile methodology. The process was appropriate for the project due to the need of constant sponsor feedback and iteration. The process was strictly followed during the course of the project. The team would choose the same process if the project was repeated, as an alternative process would have been far too rigid and strict.

2. Was there a large requirement to learn the problem domain? What approach was used to gain domain expertise? Did your sponsor provide adequate support? What forms of support did you receive?

The team allotted an entire sprint's worth of time in order to learn and understand the problem domain. As this project has been through many senior projects across multiple disciplines, there were many documents and artifacts that the team had to reverse engineer and understand. The general context was fairly easy to understand, as the team all understood the basics of hiking. The sponsor adequately supported this process, through constant discussions during meetings and through emails.

3. What mechanisms did the team use to track project progress? Did they give the team and sponsor adequate insight into project progress and issues? How well did the team track its project progress? How often did these artifacts get updated on the department project sites?

The primary tool that the team used to track progress was GitHub Projects. The team and sponsor could view a backlog, roadmap, and calendar of tasks and items that were planned, in progress, in review, and completed. The team faithfully tracked project progress, and the roadmap was constantly updated, and reviewed by the team at the beginning of each sprint. The artifacts that needed to be turned in to the project website were also tracked on the team's calendar, and were updated as needed.

4. Did the team conduct effective meetings?

Conducting effective meetings was something that the team initially struggled with and had to improve upon as the semesters progressed. Meetings were always productive, but often went too long, oftentimes taking 4+ hours, causing a lot of burnout and fatigue as the semesters progressed. The team aimed to fix this by more strictly planning out what happens in a given meeting, and moving all tasks that can be done asynchronously outside of meetings. Since the change, meetings have become much more effective and no longer stretch to the lengths that they did at the beginning of the project.

5. Did the team meet all project milestones? Which milestones, if any, were missed or were met ahead of schedule? What contributed to schedule changes? What could the team have done differently to ensure that milestones were met?

The team was able to meet every project milestone defined in the MVP. Additionally, more milestones were added to the project at the beginning of the summer semester. Factors such as scope change and lack of a device contributed to schedule changes, and required the team to adjust accordingly.

6. Was the team required to adopt new technologies? What were these technologies? What approach did the team use for selecting the appropriate technology for the project? Did the sponsor provide any support for learning these technologies? How well did the team ramp up on the new technologies and begin to apply them effectively?

Most technologies that the team utilized were familiar with at least one member of the team, but often not every member. A major example of this was AWS, where only two members of the team had prior experience working with it, while other members needed to learn it with the guidance of the members who did know it. When the team selected new technologies, usually there would be pros and cons of each option laid out, and a decision would be made in a meeting, often with the team coach present as well. The sponsor did not provide support for using these technologies as the sponsor didn't know how to use them.

7. How well did the team maintain quality control over the project artifacts? Have all artifacts been reviewed for adherence to quality standards? What was the review process used by the team?

The team was able to maintain a high quality product through vigorous testing. There is a form of testing for each stage of the project. The team constantly dedicates meeting time to review documentation and ensure that what's written is up to par. For code, the team has written unit tests (lambda tests in AWS), integration/API tests, UI tests, and manual user acceptance tests to ensure that the entire stack is high quality. Each developer is expected to create their own branch when making changes, and the repository has GitHub Actions set up to require two code reviews with an approval before anything is merged into another branch. Additionally, the team has created many testing environments, where different stages of development can be fully tested. Changes that are immediately merged from a local branch go into TST (test). Those changes are pushed to the UAT (user acceptance test), where the sponsor and other stakeholders can view changes that have been made during the previous sprint. Finally, PROD and DEMO hold a production-ready version of the application, where DEMO holds simulated/fake data to show off the application.

8. Did the team have any issues with configuration management? How were these problems solved? What percentage of project artifacts is under configuration control?

The team had little to no problems with configuration. The team spent most of early sprints setting up various configurations, such as AWS and GitHub roles. At times, the team faced issues with GitHub Actions, often due to changes made in security or AWS. These problems are solved by constant communication across the team to identify problems and fix them ASAP. Every single artifact goes through the team's configuration control in some shape or form.

9. What was the set of metrics that the team tracked? Did the team gather these metrics on a consistent basis? What did the team learn from the review of these metrics?

The team tracks time spent working each week, sprint velocity, and types of tasks completed. These tasks were consistently gathered, as time was dedicated to determining story points, and to logging hours worked each week. The team learned that implementation is only a small aspect of the project, and process-related tasks such as documentation and testing take up just as much time.

Communication and Interaction

1. How well did the team communicate project progress to the sponsor? What regular communication did the team have with the sponsor? Did the team maintain this communication to the satisfaction of the sponsor? Were any adjustments needed in the communication over time? Were these changes initiated by the team or the sponsor?

The team communicated project progress consistently to the sponsor. End-of-sprint meetings were conducted for the team to demo the previous sprint's work to the sponsor, and communication was always open through email. The sponsor was satisfied with the amount of communication throughout the project. Communications have evolved over the course of the project - as a routine was established, there was generally less need for constant communication through email, and most discussions were completed during in-person meetings.

2. Did the team need to provide technical input to the sponsor? How well did the team educate the customer in these areas? What mechanism did the team use?

Yes, we did. The sponsor had his own preconceived notions involving certain technical decisions for the product that at times were useful as a foundation, but at other times, did not align with the set goal of the project. We needed to provide input on what was feasible and what was not, along with informing the sponsor that certain changes were absolutely necessary, despite initial resistance against such.

3. Was this an effective team? What has been contributing to and detracting from the team's effectiveness? What are the team's weak points? What are the team's strong points? What changes could the team have made to make it more effective?

Yes, this team was remarkably effective. The strongest contributor is how efficient our process was, keeping everything documented and having all our tasks well defined and planned out in advance. Our biggest detractor was outside factors, as team members had additional academic commitments that were unavoidable, which occasionally could make it challenging to get work done without causing complete and total burnout, or occasionally cut planned meeting times short. The strong points were how self motivated the team members were. Team members did not frequently need to be prodded to get things done, and were self motivated to go above and beyond. The team strove to not just meet deadlines and requirements to the letter, but make a good product whose quality stood as a reflection of those who put the work into it. For how effective the team was, the only way we could have seriously improved would have simply been more time and more manpower. As both are limited resources, this team hit the point of diminishing returns on effectiveness and realistically could not have made any further significant jumps in effectivity.

4. What mechanism did the team use to communicate with the faculty coach? Was communication with the coach effective? Were there any trouble spots with the faculty coach communications? What could the team or faculty coach have changed to make their communication more effective?

The team used regular meetings and emails with the faculty coach to communicate, this model was effective. No trouble spots, the coach has been very helpful with managing the project.

5. Did the team need to interact with department staff personnel, i.e., the office staff or system administration? Was this handled in a professional manner? Were there any problems with these interactions?

Yes, our team did interact with department staff, particularly the Head of Senior Project, regarding securing a meeting space for the summer. This required consistent communication and follow-up, as there was significant back-and-forth to coordinate availability and logistics. All interactions were handled in a professional and respectful manner. We clearly communicated our needs, remained responsive, and followed appropriate channels throughout the process. Additionally, our team is going above and beyond by organizing a presentation for the Associate Dean, even though this is not a required component of the project. Overall, there were no major issues with these interactions, and we were able to successfully collaborate with department staff to meet our needs.

6. Does the team have a complete website with all project artifacts stored and up-to-date on the software engineering department webserver? How often were entries on the webserver updated?

Yes, our team has a complete and up-to-date website hosted on the software engineering department webserver. All key project artifacts, including documentation, deliverables, and relevant materials, are stored and organized there.

We updated the webserver regularly throughout the project, especially at the end of each sprint and whenever significant changes or new artifacts were produced, to ensure everything remained current and accessible. Time Tracking was made to be automated so that the table would not have to be constantly updated.

7. How well has the team made presentations to the sponsor and faculty coach? Was the final project presentation done in a professional manner? Was the poster presentation done in a professional manner? What could have been done to improve the team's presentations?

The team has made professional presentations for both the sponsor and faculty coach. The team made an end of semester presentation that was presented to the coach, sponsor, and another senior project team. The team also made a more in depth presentation and presented it to the coach, sponsor, and associate dean to elaborate on things the end of semester presentation didn't have time to cover. The presentations were practiced and timed to ensure that everything would go smoothly. There was no poster presentation due to being a summer team with no poster presentation event.

8. Does the technical report adequately document the project and its results? Was the paper of high technical and editorial (language, style, grammar, etc.) quality? Did all teammates contribute to the paper? Did the sponsor contribute to the paper? Did the sponsor review the paper?

   The technical report documents all aspects of the project and results. It was of high technical and editorial quality, with all team members contributing to different parts of the paper. Sections were outlined and assigned to each team member based on what they worked on, and was reviewed to make sure it was of superior quality.

Preparation and Resources

1. Did the team possess adequate management and process skills (team building, planning, risk management, change management, process definition and tracking, etc.) to carry out the project? If not, how could the program provide better preparation?

The team definitely possessed a high level of management and process skills. The entire team brought unique experiences from internships and coursework that allowed the team to develop great chemistry and create an extremely strong process. The program could (and should) absolutely provide more opportunities for team projects and execute the full software development lifecycle.

2. Did the team possess adequate technical skills (requirements, design, coding, testing, quality reviews, etc.) to carry out the project? If not, how could the program provide better preparation?

Overall, the technical skills of the team were very diverse, with most areas of the project having at least one team member with experience. For this project specifically, we had at least two people familiar with both React and AWS, which were the two major tools of the project that the team needed to use.

Additionally, it is worth noting that the team possessed the ability to learn new skills on their own quickly, and the self motivation to learn what they needed to learn in order to get their tasks done. Any skills that were lacked, were quickly learned by the team.

3. What technical resources (or skills, training, tools) were missing, if any, that would have helped make the senior project experience more successful?

Mobile development, in depth networking security experience, heavier experience in cloud and ci/cd development. More examples of following the entire software development lifecycle from classes. A device to work with.

4. Were the facilities adequate for you to perform your work on the project?

For the spring semester, yes. For the first few weeks of the summer semester, it was rather difficult to get a room and we had to jump through multiple hoops to track down who could give us access, then actually getting authorized for said access.

Achieving Customer Satisfaction

1. In the team's opinion did the work satisfy the project sponsor? Are there areas where you think you exceeded the sponsor's expectations? Were there any weak spots in this regard?

Overall, the team definitely satisfied the project sponsor. The sponsor has agreed that the team's MVP is complete, and everything the team set out to do at the start of the project has been completed. One area that exceeded the sponsor's expectations would be in security. Going into the project, the sponsor's expectations for security were very low, with the application being largely unsecure. With the implementation of mTLS, this expectation has far been exceeded. The biggest weak spot would probably be the heatmap. It is a requirement that was added much later into the lifecycle of the project, and immediately became a high priority when it was introduced. Due to its late addition, it is less refined than the other areas of the project, and while it is satisfactory, it acts more as a prototype than the rest of the project. There were many late additions to the requirements that necessitated many meetings to

clarify, design, and properly process. This caused requirements volatility to jump significantly and affected the team's velocity.

Achieving Team Satisfaction

1. Did the project satisfy the team's expectations for learning? What could the team have done differently to improve the team's learning experience? What could the faculty coach and department have done differently to improve the team's learning experience?

The project definitely satisfied the team's expectations for learning. Most members went into the project not really expecting to learn much, and rather apply and refine the skills we'd already learned. After completing the project, the team learned a lot about working on a project long term that was absent from most other projects that never had a lot of time dedicated to them. Things like process felt less relevant when a project would only last weeks. By working on the same project and making decisions for a lot of hours over the span of months, the team learned a lot about how to effectively plan out a project, and manage the changing requirements that come in a real world project.

Advice for Future Teams

1. What one piece, or more, of advice would you give to future senior project teams to help them be successful?

EVERYTHING is about process. Yes, it's a lot of documentation. Yes, it seems largely irrelevant when you could be writing code. But having a paper trail and a very strict structure to get things done by is incredibly important to ensuring things go smoothly. Letting everyone just "do their thing" may be the more "fun" or "lower effort" choice, but having a very strictly regimented process allows for your work to flourish without coming at the expense of the team or the individual. Documenting everything properly also reduces friction and thrashing between team members, as it can reduce the amount of miscommunication and confusion.

Another piece of advice would be to make people's role in a pull request more specific. The team wanted to ensure that all code was of acceptable quality, so all pull requests required tests to pass and at least two code reviews. One problem the team ran into was that often pull requests wouldn't be reviewed until they needed to be merged at the end of the sprint, leading to less thorough reviews and more open pull requests. It would likely be helpful to have a dedicated person to review pull requests, or at least assigning reviews to specific team members with a deadline, treating it like a regular sprint item.